

FMUSER FUQ-4513A DVB-S-S2 Modulator 1-4 ASI Ports IP Input to DVB-S-S2 RF Output BISS Scrambling EN302307 Standard for Secure Satellite Broadcasting



Product overview

The FMUSER FBE4513E is a DVB-S2 modulator that converts ASI and IP inputs to DVB-S/S2 RF output, featuring BISS scrambling for secure distribution. It processes signals by modulating them according to EN302307 standards. Controlled via Web-server NMS and front-panel LCD, it is ideal for broadcasting and interactive services.

Key feature

Complies with DVB-S2 (EN302307) and DVB-S (EN300421) standards.

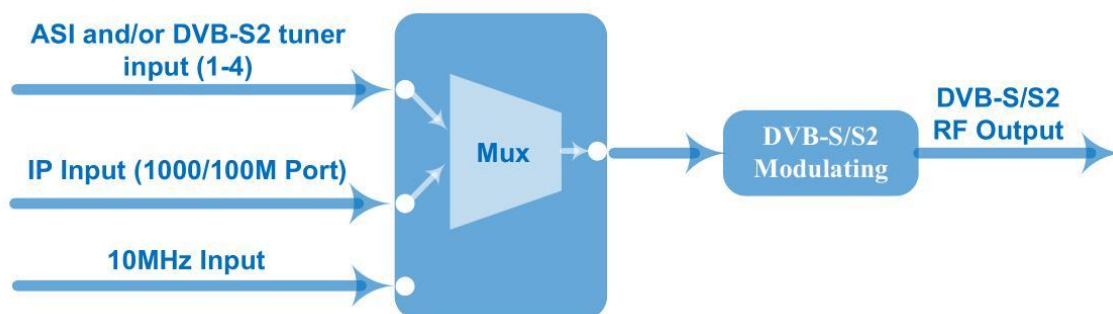
Supports ASI and IP inputs (1000/100M) for flexible integration.

Features BISS scrambling for secure program distribution.

Provides RF output at 950-2150MHz with 10KHz stepping and MER≥40dB.

Enables control via Web-server NMS and front-panel LCD.

Principle Chart



Specifications

IP Input	1*IP Input (RJ45, 1000/100M TS Over UDP)	1*IP Input (RJ45, 1000/100M TS Over UDP)	1*IP Input (RJ45, 1000/100M TS Over UDP)	1*IP Input (RJ45, 1000/100M TS Over UDP)
ASI Input	Supporting both 188/204 Byte Packet TS Input	Supporting both 188/204 Byte Packet TS Input	Supporting both 188/204 Byte Packet TS Input	Supporting both 188/204 Byte Packet TS Input
ASI Input	1-4 ASI Inputs (Connector: BNC, Impedance 75Ω)	1-4 ASI Inputs (Connector: BNC, Impedance 75Ω)	1-4 ASI Inputs (Connector: BNC, Impedance 75Ω)	1-4 ASI Inputs (Connector: BNC, Impedance 75Ω)
Tuner Input (0-4 inputs optional)				
Multiplexing	Maximum PID Remapping	Maximum PID Remapping	128 input per channel	128 input per channel
Multiplexing	Function	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Function	Function	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Function	Edit PSI/SI tables (PAT/PMT/SDT/CAT/NIT)	Edit PSI/SI tables (PAT/PMT/SDT/CAT/NIT)
10MHz Reference Clock	1*External 10MHz Input (BNC Interface); 1*Inner 10MHz Reference clock	1*External 10MHz Input (BNC Interface); 1*Inner 10MHz Reference clock	1*External 10MHz Input (BNC Interface); 1*Inner 10MHz Reference clock	1*External 10MHz Input (BNC Interface); 1*Inner 10MHz Reference clock
RF Output	RF Range: 950 ~ 2150 MHz, 10KHz stepping	RF Range: 950 ~ 2150 MHz, 10KHz stepping	RF Range: 950 ~ 2150 MHz, 10KHz stepping	RF Range: 950 ~ 2150 MHz, 10KHz stepping
RF Output	Output Level Attenuation -26 ~ 0 dBm 0.5dBm Stepping	Output Level Attenuation -26 ~ 0 dBm 0.5dBm Stepping	Output Level Attenuation -26 ~ 0 dBm 0.5dBm Stepping	Output Level Attenuation -26 ~ 0 dBm 0.5dBm Stepping
RF Output	MER ≥ 40dB	MER ≥ 40dB	MER ≥ 40dB	MER ≥ 40dB
RF Output	Connector: N type, Impedance 50Ω; N to F adapter	Connector: N type, Impedance 50Ω; N to F adapter	Connector: N type, Impedance 50Ω; N to F adapter	Connector: N type, Impedance 50Ω; N to F adapter

Channel Coding and Modulation	Standard	DVB-S	DVB-S	DVB-S2
Channel Coding and Modulation	Outer coding	RS Coding	RS Coding	BCH Coding
Channel Coding and Modulation	Inner coding	Convolution	Convolution	LDPC Coding
Channel Coding and Modulation	Constellation	QPSK	QPSK	QPSK, 8PSK, 16 APSK, 32APSK
Channel Coding and Modulation	FEC/ Convolution Rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10 32APSK: 3/4, 4/5, 5/6, 8/9, 9/10
Channel Coding and Modulation	Roll-off Factor	0.2, 0.25, 0.35	0.2, 0.25, 0.35	0.2, 0.25, 0.35
Channel Coding and Modulation	Symbol Rate	0.05~45Msps	0.05~45Msps	0.05~40Msps (32APSK); 0.05~45 Msps (16APSK/8PSK/QPSK)
BISS Scramble	Mode 0, mode 1, mode E	Mode 0, mode 1, mode E	Mode 0, mode 1, mode E	Mode 0, mode 1, mode E
System	Web-server NMS	Web-server NMS	Web-server NMS	Web-server NMS
System	Language: English	Language: English	Language: English	Language: English
System	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
System	24V power output through RF output port	24V power output through RF output port	24V power output through RF output port	24V power output through RF output port
Miscellaneous	Dimension	482mm×410mm×44mm	482mm×410mm×44mm	482mm×410mm×44mm
Miscellaneous	Temperature	0~45℃(operation), -20~80℃(storage)	0~45℃(operation), -20~80℃(storage)	0~45℃(operation), -20~80℃(storage)

Miscellaneous	Power	100-240VAC±10 %, 50Hz-60Hz	100-240VAC±10 %, 50Hz-60Hz	100-240VAC±10 %, 50Hz-60Hz
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Order Guide

	FU-4513E (New version)	FU-4513E (Old version)	FU-4513F	FU-4514	FU-4514F
Input	1 IP (100/1000M) input, ASI and/or DVB-S2 tuner inputs for re-mux or pass through	1 IP (100M) or 1 ASI pass through alternative	1 IP (100M) or 1 ASI pass through alternative	1 IP (100M) or 1 ASI pass through alternative	1 IP (100M) or 1 ASI pass through alternative
DVB-S/S2 out	●	●	●	●	●
DVB- S2X out				●	●
QPSK, 8PSK	●	●	●	●	●
16APSK, 32APSK	●	●		●	●
8APSK-L,16A PSK-L,32APS K-L				●	●
RF output (95 0-2150MHz)	●	●		●	
IF output (50-960MHz)			●		●
Mux	●				

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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